COMPUTER PROGRAMMING

Assignment No. 2 · Functions | 1D Arrays | 2D Arrays | String Arrays

Student Name	Kareem Ullah
Registration No.	BF25NWELE0721
Section	Electrical Power
Semester	2nd Semester
Subject	Computer Programming
Submitted To	Engr. M. Rizwan
Submission Date	4th May, 2026
University	UET Peshawar — Nowshera Campus

Q1. 1D Array — Maximum, Minimum & Average

```
#include <iostream>
using namespace std;

int main() {     int arr[10], sum =
0, max, min;

    // input 10 integers     cout
<< "Enter 10 integers: ";     for
(int i = 0; i < 10; i++)
cin >> arr[i];

    max = min = arr[0];  // start comparison from first element

    for (int i = 0; i < 10; i++) {
if (arr[i] > max) max = arr[i];
if (arr[i] < min) min = arr[i];
sum += arr[i];
    }

    // display results     cout << "Max: "
<< max         << endl;     cout << "Min: "
<< min         << endl;     cout << "Average: "
<< sum / 10.0  << endl;

    return 0;
}
```

Q2. 1D Array — Reverse Using a Function

```
#include <iostream>
using namespace std;

// reverses array in-place using two-pointer swap
void reverseArray(int arr[], int n) {     int
temp;     for (int i = 0; i < n / 2; i++) {
temp          = arr[i];         arr[i]        =
arr[n - 1 - i];         arr[n - 1 - i] = temp;
```

```
    }
}

void printArray(int arr[], int n)
{     for (int i = 0; i < n; i++)
cout << arr[i] << " ";     cout <<
endl;
}

int main() {
    int arr[] = {10, 20, 30, 40, 50};
int n = 5;

    cout << "Before: "; printArray(arr, n);
reverseArray(arr, n);                // call reverse function
cout << "After:  "; printArray(arr, n);

    return 0;
}
```

Q3. 2D Array — 3×3 Matrix Addition

```
#include <iostream>
using namespace std;

int main() {     int A[3][3],
B[3][3], C[3][3];

    // input first matrix     cout << "Enter
elements of Matrix A:" << endl;     for (int i =
0; i < 3; i++)         for (int j = 0; j < 3; j++)
cin >> A[i][j];

    // input second matrix     cout << "Enter
elements of Matrix B:" << endl;     for (int i =
0; i < 3; i++)         for (int j = 0; j < 3; j++)
cin >> B[i][j];

    // add corresponding elements
for (int i = 0; i < 3; i++)         for
(int j = 0; j < 3; j++)
C[i][j] = A[i][j] + B[i][j];

    // display result matrix     cout
<< "Result Matrix:" << endl;     for
(int i = 0; i < 3; i++) {         for
(int j = 0; j < 3; j++)
cout << C[i][j] << " ";         cout
<< endl;
    }

    return 0;
}
```

Q4. String Array — Alphabetical Sorting

```
#include <iostream> #include
<string>
using namespace std;


int main() {     string
names[5], temp;


    // input 5 names     cout
<< "Enter 5 names: ";     for
(int i = 0; i < 5; i++)
cin >> names[i];
```

```


    // selection sort — swap if out of alphabetical order
for (int i = 0; i < 5; i++)         for (int j = i + 1; j
< 5; j++)             if (names[i] > names[j]) {
temp     = names[i];                 names[i] = names[j];
names[j] = temp;
            }


    // display sorted names
cout << "Sorted: ";     for
(int i = 0; i < 5; i++)
        cout << names[i] << " ";


    return 0;
}
```

Q5. Power System Load Analysis

```
#include <iostream>
using namespace std;


// returns sum of all hourly loads float
totalLoad(float loads[], int n) {     float sum
= 0;     for (int i = 0; i < n; i++) sum +=
loads[i];     return sum;
}


// returns highest load value float
peakLoad(float loads[], int n) {     float
peak = loads[0];     for (int i = 1; i < n;
i++)         if (loads[i] > peak) peak =
loads[i];     return peak;
}


int main() {
    float loads[24];


    // input 24-hour load data     cout << "Enter load
demand for 24 hours (MW):" << endl;     for (int i = 0; i <
24; i++) {         cout << "Hour " << i + 1 << ": ";
cin >> loads[i];
    }


    cout << "Total Load : " << totalLoad(loads, 24) << " MW" << endl;
cout << "Peak Load  : " << peakLoad(loads, 24)  << " MW" << endl;


    return 0;
}
```

Q6. Student Result System

```
#include <iostream> #include
<string>
using namespace std;

const int N = 5;

// finds and prints student with highest marks
void findTopper(string names[], int marks[]) {
int top = 0;
    for (int i = 1; i < N; i++)
        if (marks[i] > marks[top]) top = i;
    cout << "Topper: " << names[top] << " (" << marks[top] << ")" << endl;
}
```

```
// prints pass/fail for each student (pass >= 50)
void showResult(string names[], int marks[]) {
for (int i = 0; i < N; i++)
        cout << names[i] << ": " << (marks[i] >= 50 ? "Pass" : "Fail") << endl;
}

// searches student by name and prints marks void searchStudent(string
names[], int marks[]) {     string name;     cout << "Enter name to search: ";
cin >> name;     for (int i = 0; i < N; i++)         if (names[i] == name) {
cout << "Found: " << names[i] << " - Marks: " << marks[i] << endl;
return;
        }
    cout << "Not found." << endl;
}

int main() {
string names[N];
    int marks[N];

    for (int i = 0; i < N; i++) {
cout << "Name: ";  cin >> names[i];
cout << "Marks: "; cin >> marks[i];
    }

    findTopper(names, marks);
showResult(names, marks);
    searchStudent(names, marks);

    return 0;
}
```

Q7. Renewable Energy Monitoring System

```
#include <iostream>
using namespace std;

// returns total energy generated float
calcTotal(float e[], int n) {     float sum
= 0;     for (int i = 0; i < n; i++) sum +=
e[i];     return sum;
}

// returns index of hour with max
generation int findPeakHour(float e[], int
n) {     int peak = 0;
    for (int i = 1; i < n; i++)
if (e[i] > e[peak]) peak = i;
return peak;
}

// returns average energy per hour float
calcAverage(float total, int n) {
return total / n;
}

int main() {
    float energy[24];

    // input hourly energy data
    cout << "Enter energy generated each hour (kWh):" <<
endl;     for (int i = 0; i < 24; i++) {         cout << "Hour
" << i + 1 << ": ";         cin >> energy[i];
    }

    float total = calcTotal(energy, 24);

    int   peak  = findPeakHour(energy, 24);

    cout << "Total Energy : " << total                    << " kWh" << endl;
cout << "Average      : " << calcAverage(total, 24)   << " kWh" << endl;
    cout << "Peak Hour    : Hour " << peak + 1 << " (" << energy[peak] << " kWh)" << endl;

    return 0;
}
```

Q8. Battery Health Monitoring

```
#include <iostream>
using namespace std;

// counts readings outside safe range [3.0V, 4.2V] and
warns void checkSafety(float v[], int n) {     int below =
0, above = 0;     for (int i = 0; i < n; i++) {         if
(v[i] < 3.0) below++;         else if (v[i] > 4.2) above++;
    }     cout << "Below 3.0V : " << below <<
endl;     cout << "Above 4.2V : " << above <<
endl;     if (below > 0 || above > 0)
        cout << "WARNING: Unsafe readings detected!" << endl;
}

// finds and prints minimum and maximum
voltage void findMinMax(float v[], int n) {
float mn = v[0], mx = v[0];     for (int i = 1;
i < n; i++) {         if (v[i] < mn) mn = v[i];
if (v[i] > mx) mx = v[i];
    }     cout << "Min Voltage : " << mn << " V" <<
endl;     cout << "Max Voltage : " << mx << " V" <<
endl;
}

int main() {     float
voltages[20];

    // input 20 voltage readings     cout << "Enter
20 voltage readings (V):" << endl;     for (int i =
0; i < 20; i++) {         cout << "Reading " << i + 1
<< ": ";         cin >> voltages[i];
    }

    checkSafety(voltages, 20);
    findMinMax(voltages, 20);

    return 0;
}
```

Q9. Smart Grid Load Distribution

```
#include <iostream>
using namespace std;

const int R = 4, D = 7;  // 4 regions, 7 days

// prints total consumption for each
region void regionTotals(float g[R][D]) {
for (int r = 0; r < R; r++) {         float
sum = 0;
        for (int d = 0; d < D; d++) sum += g[r][d];
        cout << "Region " << r + 1 << " Total: " << sum << " MW" <<
endl;     } }
```

```
// finds day with highest combined demand across all
regions void peakDay(float g[R][D]) {     float daySum[D] =
{0};     for (int d = 0; d < D; d++)         for (int r = 0;
r < R; r++)             daySum[d] += g[r][d];     int peak =
0;     for (int i = 1; i < D; i++)         if (daySum[i] >
daySum[peak]) peak = i;
    cout << "Peak Day: Day " << peak + 1 << " (" << daySum[peak] << " MW)" << endl;
}

// returns average over all regions and
days float overallAvg(float g[R][D]) {
float sum = 0;     for (int r = 0; r < R;
r++)         for (int d = 0; d < D; d++)
sum += g[r][d];
    return sum / (R * D);
}

int main() {
    float grid[R][D];

    cout << "Enter consumption for 4 regions over 7 days (MW):" <<
endl;     for (int r = 0; r < R; r++)         for (int d = 0; d < D;
d++) {             cout << "Region " << r+1 << " Day " << d+1 << ": ";
cin >> grid[r][d];
        }

    regionTotals(grid);
peakDay(grid);
    cout << "Overall Average: " << overallAvg(grid) << " MW" << endl;

    return 0;
}
```

Q10. Classroom Attendance System

```
#include <iostream>
using namespace std;

const int S = 5, D = 7;  // 5 students, 7 days

// prints total days present for each
student void studentAttendance(int a[S][D]) {
for (int s = 0; s < S; s++) {         int
total = 0;
        for (int d = 0; d < D; d++) total += a[s][d];         cout << "Student "
<< s + 1 << ": " << total << " days present" << endl;     }
}

// finds student with highest attendance void
bestStudent(int a[S][D]) {     int best = 0,
bestCount = 0;     for (int s = 0; s < S; s++) {
int count = 0;         for (int d = 0; d < D; d++)
count += a[s][d];
        if (count > bestCount) { bestCount = count; best = s; }
    }
    cout << "Highest Attendance: Student " << best + 1 << " (" << bestCount << " days)" << endl;
}

// calculates overall class attendance
percentage void classPercentage(int a[S][D]) {
int present = 0;
    for (int s = 0; s < S; s++)
for (int d = 0; d < D; d++)
```

```
            present += a[s][d];
    cout << "Class Attendance: " << (present * 100.0) / (S * D) << "%" << endl;
}

int main() {
    int att[S][D];

    // input attendance: 1 = present, 0 = absent     cout <<
"Enter attendance (1=Present, 0=Absent):" << endl;     for
(int s = 0; s < S; s++) {         cout << "Student " << s + 1
<< " (7 days): ";         for (int d = 0; d < D; d++) cin >>
att[s][d];
    }

    studentAttendance(att);
bestStudent(att);
    classPercentage(att);

    return 0;
}
```